

Cofinal-orbit rigidity at the ultraexacting boundary

Sharp cardinal-size barriers for definable multisections,
quotient homomorphisms, and preserved enrichments

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Abstract. Let λ be ultraexacting, let D_λ consist of the cofinal subsets of λ of order type ω , and identify two members when their symmetric difference is finite. We prove that there is one equivalence class q such that every nonempty family of short cofinal subsets of λ definable from q , ordinal parameters, and finitely many parameters in V_λ has cardinality at least λ . The bound is attained by q itself. Consequently every complete section definable from these parameters meets this *same* class in exactly λ members. The conclusion also holds simultaneously for every member of a definable family of fewer than λ complete sections, without requiring its members to be individually definable.

A second consequence excludes definable small families of homomorphisms from finite-difference equivalence to any definable equivalence relation on D_λ all of whose classes have cardinality below λ . No uniform bound on these class sizes is assumed. We also give a direct obstruction to an Icarus-type self-embedding preserving a thin complete section. The principal proof isolates a small-fixed-family lemma and a uniform minimal-rank parameter argument; these avoid both a false singular-cardinal union estimate and an unjustified assumption that arbitrary ordinal parameters are fixed.

The conditional inconsistency obtained

Writing $[a]$ for the finite-difference class of a , we prove

$$\text{ZFC} \vdash \text{UEX}(\lambda) \implies \neg \exists T \in \text{OD}_{V_\lambda} \left[T \subseteq D_\lambda \wedge \forall a \in D_\lambda \left(0 < |T \cap [a]| < \lambda \right) \right].$$

The stronger simultaneous and parameter-relative statements are in Theorems 5.3, 6.1, and 6.4. All new deductions are proved in detail. The known I0 equiconsistency supplies a consistency calibration, not a new comparison between standard large-cardinal axioms.

Status. This is an unrefereed mathematical continuation, not a machine-checked proof. The assertions are stronger than the corresponding statements in the supplied synthesis; published priority is not claimed. No unconditional inconsistency of ultraexacting cardinals, I0, or another standard large-cardinal hypothesis is asserted.

Contents

1	Research delta and scope	3
2	Setting, notation, and external inputs	4
3	Local cofinal rigidity	5
4	Uniform control of low-rank parameters	6
5	A cofinal-rigid tail-equivalence class	8
6	Sharp multisection and trace barriers	9
7	Non-compression of quotient equivalence	11
8	A direct preserved-enrichment inconsistency	12
9	Consistency calibration and explicit nonclaims	13
10	Further questions and a usable test	14
A	Definability, reflection, and the parameter minimum	14
B	Rank bookkeeping and proof audit	15
C	Relation to the supplied reports and literature	17

Reading map

Section 3 proves the local obstruction, Section 4 handles all low-rank parameters simultaneously, and Section 5 constructs the distinguished tail-equivalence class. Sections 6–8 give the conditional inconsistency results. Section 9 separates the new deductions from imported consistency results. The appendices record the definability construction and a proof audit.

1 Research delta and scope

The supplied archive contains the *Unified Report* and the subsequent *Synthesis of the nine continuation reports* [1, 2]. The latter already proves a lower bound of λ for definable families of short cofinal maps at an exacting cardinal, and a finite-valued-transversal obstruction at an ultraexacting cardinal. It also treats finite families of selectors on a finite-subset quotient. The present continuation does not relabel those results as new.

The extension here has three distinct features. First, “finite nonzero” is replaced by “nonzero and strictly below λ ,” even with no common upper bound below λ . Second, one quotient class works *simultaneously* for all the relevant definable sets; the class itself may then be used as a parameter. Third, the argument treats small families whose individual members need not be definable, and gives a non-compression theorem for equivalence-relation homomorphisms. These strengthenings depend on the local argument developed below, not on the forcing constructions in the synthesis.

Question	Previous synthesis	This continuation
Complete sections of D_λ/E_0	No definable section with every intersection finite and nonempty.	No section with every intersection nonempty and $< \lambda$; a common full-size class exists.
Low-rank parameters	A witness is arranged to fix the particular parameters needed for a counterexample.	Minimal parameter rank yields one class with rigidity over <i>all</i> low-rank parameters, and over the class itself.
Families of sections	Finite-symmetry arguments provide related finite-family obstructions.	Every member of a definable family of size $< \lambda$ has empty or full-size trace on the distinguished class.
Quotient maps	Not the homomorphism obstruction proved here.	No definable small family of homomorphisms into relations with all classes $< \lambda$.

These are conditional inconsistency results: an ultraexacting cardinal cannot coexist with the specified *definability* or *preservation* principles. Ordinary Choice is not one of the forbidden principles. It still supplies nondefinable transversals.

The finite-subset selector question in the synthesis is different: a function choosing an element of a finite set of quotient classes is not a complete section choosing representatives *inside* those classes. In particular, this report does not settle whether there can be a countable definable family of binary selectors on Q_λ .

2 Setting, notation, and external inputs

2.1 Ambient theory and the quotient

We work in $V \models \text{ZFC}$. All cardinalities, unless explicitly qualified, are computed in V . The symbol λ denotes an uncountable limit cardinal. Define

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega \text{ and } \sup a = \lambda\}, \quad (2.1)$$

$$a E_0 b \iff |a \triangle b| < \omega, \quad (a, b \in D_\lambda), \quad (2.2)$$

$$Q_\lambda = D_\lambda / E_0, \quad [a] = \{b \in D_\lambda : b E_0 a\}, \quad (2.3)$$

$$C_\lambda = \{u \subseteq \lambda : \sup u = \lambda \text{ and } \text{ot}(u) < \lambda\}. \quad (2.4)$$

A member of C_λ is called a *short cofinal set*. Since λ is a cardinal and $u \subseteq \lambda$,

$$\text{ot}(u) < \lambda \iff |u| < \lambda. \quad (2.5)$$

Indeed $\text{ot}(u) \leq \lambda$, and equality is equivalent to its cardinality being λ . Notice that $D_\lambda \subseteq C_\lambda$ whenever $\text{cf}(\lambda) = \omega$.

Lemma 2.1 (Size of a tail-equivalence class). If $a \in D_\lambda$, then $|[a]| = \lambda$.

Proof. Every $b \in [a]$ is obtained from a by deleting finitely many elements and adding finitely many ordinals below λ . The number of pairs of such finite sets is at most λ . Conversely, for each $\beta \in \lambda \setminus a$, the set $a \cup \{\beta\}$ belongs to $[a]$. These sets are distinct and there are λ of them. Inserting finitely many ordinals into a cofinal increasing ω -sequence still gives order type ω : below each inserted ordinal there were only finitely many terms of the original sequence. $\square$

2.2 Precisely which parameters are allowed

For fixed sets p, r , let $\text{OD}(p, r)$ be the class of sets ordinal definable with p and r as additional parameters. Define

$$\text{OD}_{V_\lambda, r} = \bigcup_{p \in V_\lambda} \text{OD}(p, r), \quad \text{OD}_{V_\lambda} = \bigcup_{p \in V_\lambda} \text{OD}(p). \quad (2.6)$$

Finitely many parameters from V_λ can be combined into one such p . The notation $\text{OD}_{V_\lambda, q}$ means that q is allowed as *one set parameter*; it does *not* allow an arbitrary member of q as a parameter. This distinction is indispensable: a member of q is itself a short cofinal set.

We use the standard, uniformly definable, set-like well-order of $\text{OD}(p, r)$. A construction and the required reflection justification are given in Appendix A. No well-order of V_λ is presumed to be ordinal definable.

2.3 Exact and ultraexact witnesses

An exact witness at λ , at a sufficiently correct height $\zeta > \lambda$, has the form

$$X < V_\zeta, \quad V_\lambda \cup \{\lambda\} \subseteq X, \quad j : X \longrightarrow V_\zeta, \quad j(\lambda) = \lambda, \quad \text{crit}(j) < \lambda. \quad (2.7)$$

The map is elementary, and X is not assumed transitive. A witness is *ultraexact* when it also satisfies

$$e = j \upharpoonright V_\lambda \in X. \quad (2.8)$$

A cardinal is ultraexacting when such witnesses are available at arbitrarily high sufficiently correct heights; we use the equivalent standard characterizations from [3, 4]. Let $C^{(m)}$ denote the ordinals ζ for which $V_\zeta <_{\Sigma_m} V$.

Input A (Local Kunen inconsistency). In ZFC, there is no nontrivial elementary self-embedding of $V_{\mu+2}$ at a limit ordinal μ . We use this set-sized form of Kunen's theorem [5, 6].

Input B (Availability of witnesses). If λ is ultraexacting, an ultraexact witness (2.7) exists at every sufficiently high cardinal $\zeta \in C^{(2)}$. In particular, the height may be chosen in $C^{(m)}$ for any prescribed finite $m \geq 2$. See [4, Lemma 2.2 and Proposition 2.4], with the underlying characterizations in [3].

Input C (Consistency calibration). Over ZFC, the existence of an ultraexacting cardinal is equiconsistent with I0. Moreover, an ultraexacting λ can be preserved while forcing $V_\lambda \subseteq \text{HOD}$. These are respectively Theorem A and Proposition 3.9 of [4].

Only Inputs A and B enter the new obstruction proofs. Input C is used exclusively in Section 9. Nothing here depends on a claimed preservation of strong compactness by a Prikry construction, on a cover-exacting upper bound, or on an unverified Icarus-iteration theorem.

3 Local cofinal rigidity

3.1 Where the critical sequence must end

Lemma 3.1 (Normalization and fixed ordinals). Let j satisfy (2.7), put $e = j \upharpoonright V_\lambda$, and let $\kappa = \text{crit}(j)$. Then $e : V_\lambda \rightarrow V_\lambda$ is elementary. For

$$\kappa_0 = \kappa, \quad \kappa_{n+1} = e(\kappa_n),$$

we have

$$\sup_{n < \omega} \kappa_n = \lambda, \quad \kappa \leq \xi < \lambda \implies j(\xi) > \xi. \quad (3.1)$$

In particular, $\text{cf}^V(\lambda) = \omega$. Every fixed ordinal below λ is below κ , and every element of V_κ is fixed by j .

Proof. The set V_λ belongs to X , is fixed by j , and all its elements belong to X . Relativizing formulas to this set shows that e is elementary; this is valid despite the possible nontransitivity of X .

The critical point is a nonzero limit ordinal: a least moved successor would have a moved predecessor. Elementary embeddings are increasing on ordinals, so $\kappa < e(\kappa)$ and the critical sequence is strictly increasing. Let $\mu = \sup_n \kappa_n \leq \lambda$. If $\mu < \lambda$, the sequence belongs to V_λ and e sends it to its shift. Hence $e(\mu) = \mu$. Since $\mu + 2 < \lambda$, the restriction of e to $V_{\mu+2}$ is then a nontrivial elementary self-embedding, contrary to Input A. Thus $\mu = \lambda$.

For $\kappa \leq \xi < \lambda$, choose n with $\kappa_n \leq \xi < \kappa_{n+1}$. Monotonicity gives

$$j(\xi) = e(\xi) \geq e(\kappa_n) = \kappa_{n+1} > \xi.$$

This proves the assertion about fixed ordinals. Finally, rank induction proves $j(x) = x$ for $x \in V_\kappa$. At each stage $\alpha < \kappa$, both the rank bound and every possible element of that rank have already been fixed. Elementarity therefore gives equality of the sets, not merely inclusion of their pointwise images. $\square$

Remark 3.2. In proving normalization, the critical sequence is initially formed *externally*. The contradiction when its supremum is below λ puts it inside V_λ . If its supremum is λ , an ordinary exact witness need not contain the sequence. That is why ultraexactness is needed later.

3.2 A fixed short cofinal set is impossible

Lemma 3.3 (No fixed short cofinal set). For a witness as in (2.7), there is no $u \in X \cap C_\lambda$ with $j(u) = u$.

Proof. Suppose such u exists. The ordinal $\tau = \text{ot}(u)$ and the unique increasing enumeration $h : \tau \rightarrow u$ belong to X : they are uniquely definable from u in V_ζ . Their definitions are absolute at the chosen height. Since $j(u) = u$, uniqueness gives $j(\tau) = \tau$ and $j(h) = h$.

Now $\tau < \lambda$, so Lemma 3.1 gives $\tau < \kappa$. For every $\xi < \tau$, all parameters needed for evaluating $h(\xi)$ belong to X , and

$$j(h(\xi)) = j(h)(j(\xi)) = h(\xi).$$

Again by Lemma 3.1, $h(\xi) < \kappa$. Consequently $u \subseteq \kappa$, contradicting $\sup u = \lambda$. $\square$

3.3 The small-fixed-family lemma

Lemma 3.4 (No small fixed family of short cofinal sets). For a witness as in (2.7), there is no set $\mathcal{A} \in X$ satisfying

$$j(\mathcal{A}) = \mathcal{A}, \quad \emptyset \neq \mathcal{A} \subseteq C_\lambda, \quad |\mathcal{A}| < \lambda. \quad (3.2)$$

There is no assumption that the members of $\mathcal{A}$ have a common size bound below λ .

Proof. Assume (3.2). Form the order-type spectrum

$$S = \{\text{ot}(u) : u \in \mathcal{A}\} \subseteq \lambda.$$

This set belongs to X and is fixed by j , because it is uniquely definable from $\mathcal{A}$. Also $|S| \leq |\mathcal{A}| < \lambda$.

If S were cofinal in λ , equation (2.5) would put S in C_λ , contradicting Lemma 3.3. Thus S is bounded in λ . Choose an infinite ordinal $\nu < \lambda$ that bounds all its members. With $\chi = |\mathcal{A}| < \lambda$, ambient Choice yields

$$\left| \bigcup \mathcal{A} \right| \leq \chi \cdot |\nu| \leq \max(\chi, |\nu|, \aleph_0) < \lambda. \quad (3.3)$$

The union is cofinal because $\mathcal{A}$ is nonempty and every member is cofinal. The union belongs to X and is fixed by j , since it is the uniquely defined union of the fixed set $\mathcal{A}$. Hence it is a member of $X \cap C_\lambda$ fixed by j , again contradicting Lemma 3.3. $\square$

Why the singular-cardinal step is substantive

At this point $\text{cf}(\lambda) = \omega$. The inference “a union of fewer than λ sets of size below λ has size below λ ” is false in general. For example, a cofinal sequence of cardinals below λ , regarded as sets, has union λ . In Lemma 3.4, the *fixed order-type spectrum* first forces a single bound $\nu < \lambda$. Only then is the union estimate valid. The proof never assumes that j fixes each member of $\mathcal{A}$, or that it has finite orbits on $\mathcal{A}$.

4 Uniform control of low-rank parameters

The next lemma is stronger than arranging a high critical point for one prescribed parameter. It treats every parameter in V_λ at once, while allowing one arbitrary fixed set parameter r .

Write

$$\text{Good}_\lambda(A) \iff \emptyset \neq A \subseteq C_\lambda \text{ and } |A| < \lambda. \quad (4.1)$$

Both this property and membership in $\text{OD}(p, r)$ are first-order definable uniformly in their arguments. Let $\Psi(p, r, \lambda)$ assert

$$\exists A (A \in \text{OD}(p, r) \wedge \text{Good}_\lambda(A)).$$

Also fix a formula saying that A is the least such object in the canonical $\text{OD}(p, r)$ well-order. Choose a finite $m \geq 2$ large enough that these formulas and their subformulas are absolute between V and V_ζ whenever $\zeta \in C^{(m)}$. This is one fixed choice of m , not a separate choice for each definition of A ; ordinal definability has a uniform first-order definition.

Lemma 4.1 (Fixed-parameter rigidity). Let $\zeta \in C^{(m)}$ be a sufficiently high cardinal, let j satisfy (2.7), and let $r \in X$ satisfy $j(r) = r$. Then

$$A \in \text{OD}_{V_\lambda, r}, \quad \emptyset \neq A \subseteq C_\lambda \implies |A| \geq \lambda. \quad (4.2)$$

Proof. Suppose (4.2) fails. Define the set of possible low-rank parameters

$$I = \{p \in V_\lambda : \Psi(p, r, \lambda)\}.$$

It is nonempty. By the uniform reflection choice, V_ζ computes this same set I . It is uniquely definable there from r and λ , so $I \in X$ and $j(I) = I$.

Let

$$\rho = \min\{\text{rank}(p) : p \in I\}.$$

The ordinal $\rho < \lambda$ belongs to X and is fixed by j . Lemma 3.1 therefore gives $\rho < \kappa$. Choose any actual $p \in I$ of rank ρ . No definable choice of this p is needed: because κ is a limit ordinal, $p \in V_\kappa \subseteq X$, and consequently $j(p) = p$.

Now let A_* be the canonical least member of $\text{OD}(p, r)$ satisfying Good_λ . All candidate sets have rank bounded by $\lambda + 2$, so they lie in V_ζ . The defining formula for A_* reflects correctly by our choice of m . Thus A_* is uniquely definable in V_ζ from the fixed parameters p, r, λ . Elementarity yields

$$A_* \in X, \quad j(A_*) = A_*.$$

This contradicts Lemma 3.4. □

Remark 4.2 (What is and is not fixed). The original family A need not belong to X , and its original ordinal parameters need not be fixed. The proof first minimizes *the rank of a possible low-rank parameter*, and then minimizes the counterexample in the corresponding canonical definable well-order. Only that selected counterexample is shown to be fixed. Nor is I pointwise fixed a priori; its least occurring rank is fixed, which suffices.

Corollary 4.3 (Relative cofinal-map barrier). Under the hypotheses of Lemma 4.1, there is no cofinal map $f : \mu \rightarrow \lambda$ with $\mu < \lambda$ and $f \in \text{OD}_{V_\lambda, r}$. More generally, every nonempty $\text{OD}_{V_\lambda, r}$ family of such maps, with possibly varying domains below λ , has size at least λ .

Proof. For a single map use $A = \{\text{ran}(f)\}$. For a family $\mathcal{F}$ use $A = \{\text{ran}(f) : f \in \mathcal{F}\}$. The set A is nonempty, belongs to the same definability class, consists of short cofinal sets, and has cardinality at most $|\mathcal{F}|$. Apply Lemma 4.1. □

5 A cofinal-rigid tail-equivalence class

Lemma 5.1 (An internal critical orbit). If j is ultraexact, its critical sequence $c = \langle \kappa_n : n < \omega \rangle$ belongs to X . With $a = \text{ran}(c)$ and $q = [a]$, we have

$$a \in X \cap D_\lambda, \quad q \in X \cap Q_\lambda, \quad j(a) = a \setminus \{\kappa_0\}, \quad j(q) = q. \quad (5.1)$$

Proof. By ultraexactness, $e = j \upharpoonright V_\lambda$ is an element of X . Its critical point and the result of the ω -recursion $\kappa_{n+1} = e(\kappa_n)$ are uniquely definable from e in V_ζ . The sequence has rank λ and lies in V_ζ . Elementarity therefore puts c and a in X . Lemma 3.1 gives $a \in D_\lambda$.

Every $n < \omega$ is fixed, so

$$j(c)(n) = j(c(n)) = j(\kappa_n) = \kappa_{n+1}.$$

Thus $j(a)$ is exactly the tail displayed in (5.1). We have not assumed $j(e) = e$.

The sets D_λ , E_0 , and Q_λ are definable from λ and are correctly computed in V_ζ ; hence they belong to X and are fixed. The class $q = [a]$ belongs to X , and

$$j(q) = [j(a)] = [a] = q,$$

since a and $j(a)$ differ by one element. □

Definition 5.2 (Cofinal rigidity). For $q \in Q_\lambda$, write $\text{CR}(\lambda, q)$ when

$$\forall A \in \text{OD}_{V_\lambda; q} \quad [\emptyset \neq A \subseteq C_\lambda \implies |A| \geq \lambda]. \quad (5.2)$$

This is a property of a set parameter, not a newly proposed large-cardinal axiom.

Theorem 5.3 (Cofinal-orbit rigidity). If λ is ultraextending, there is $q \in Q_\lambda$ such that $\text{CR}(\lambda, q)$. In fact, the bound is sharp:

$$\min\{|A| : \emptyset \neq A \subseteq C_\lambda, A \in \text{OD}_{V_\lambda; q}\} = \lambda. \quad (5.3)$$

The same minimum is obtained when $A \subseteq C_\lambda$ is replaced by $A \subseteq D_\lambda$.

Proof. Choose the finite m from Section 4. By Input B, choose an ultraexact witness at a cardinal $\zeta \in C^{(m)}$ high enough for the preceding constructions. Lemma 5.1 supplies $q \in X \cap Q_\lambda$ with $j(q) = q$. Apply Lemma 4.1 with $r = q$ to obtain (5.2).

For sharpness, q itself is a nonempty subset of $D_\lambda \subseteq C_\lambda$, is definable from the parameter q , and has size λ by Lemma 2.1. Thus it attains both minima. □

The simultaneous quantifier is essential

Theorem 5.3 chooses q before quantifying over definable families. Its form is

$$\exists q \in Q_\lambda \forall p \in V_\lambda \forall A \in \text{OD}(p, q) \quad (\text{Good}_\lambda(A) \text{ is false}),$$

not merely a statement with a different q for every p or A . The set parameter q carries the entire tail class, but it still does not definably provide even one short cofinal set.

Corollary 5.4 (Invariant cofinal summaries). Assume $\text{CR}(\lambda, q)$. If $R \in \text{OD}_{V_\lambda; q}$ is a function on Q_λ with $R(s) \subseteq C_\lambda$ for every $s \in Q_\lambda$, then

$$R(q) = \emptyset \quad \text{or} \quad |R(q)| \geq \lambda. \quad (5.4)$$

Consequently there is no $R \in \text{OD}_{V_\lambda}$ on Q_λ with $\emptyset \neq R(s) \subseteq C_\lambda$ and $|R(s)| < \lambda$ for every $s \in Q_\lambda$.

Proof. Function evaluation makes $R(q)$ a member of $\text{OD}_{V_\lambda; q}$. Equation (5.4) is exactly (5.2). For the last assertion choose q by Theorem 5.3 and use $\text{OD}_{V_\lambda} \subseteq \text{OD}_{V_\lambda; q}$. $\square$

Equivalently, no definable function on D_λ can assign a nonempty $< \lambda$ -sized family of short cofinal sets while being constant on every E_0 -class. Such a function descends to the quotient without choosing representatives: its value at $s \in Q_\lambda$ is the unique value shared by all members of s . The identity map on Q_λ , viewed as taking values in $\mathcal{P}(C_\lambda)$, shows why $< \lambda$ cannot be replaced by $\leq \lambda$.

6 Sharp multisection and trace barriers

6.1 One class works for all definable subsets

Theorem 6.1 (Empty-or-full-size trace). Assume $\text{CR}(\lambda, q)$. For every $T \in \text{OD}_{V_\lambda; q}$ with $T \subseteq D_\lambda$,

$$|T \cap q| \in \{0, \lambda\}. \quad (6.1)$$

In particular, at an ultraexacting λ there is one q for which (6.1) holds simultaneously for all $T \in \text{OD}_{V_\lambda}$, and even for all $T \in \text{OD}_{V_\lambda; q}$.

Proof. The trace $T \cap q$ is definable from T and q , hence belongs to $\text{OD}_{V_\lambda; q}$. It is a family of members of $D_\lambda \subseteq C_\lambda$. If nonempty, cofinal rigidity gives $|T \cap q| \geq \lambda$, whereas $T \cap q \subseteq q$ and $|q| = \lambda$ give the reverse inequality. $\square$

Definition 6.2 (Complete and thin sections). A set $T \subseteq D_\lambda$ is a *complete section* when $T \cap s \neq \emptyset$ for all $s \in Q_\lambda$. It is λ -*thin* when additionally $|T \cap s| < \lambda$ for all $s \in Q_\lambda$. A transversal has exactly one member in each class and is therefore λ -thin.

Corollary 6.3 (Conditional inconsistency of thin definable sections). If λ is ultraexacting, no λ -thin complete section belongs to OD_{V_λ} . More precisely, there is $q \in Q_\lambda$ such that

$$T \in \text{OD}_{V_\lambda; q} \text{ and } T \text{ a complete section} \implies |T \cap q| = \lambda.$$

Proof. Choose q by Theorem 5.3. A complete section has nonempty trace on q , so Theorem 6.1 forces that trace to have size λ . $\square$

This proves the conditional inconsistency stated on the first page. It permits the forbidden section to have, for example, countably many representatives in one class, $\aleph_7$ in another, and sizes unbounded in λ across the quotient. The absence of a common bound does not weaken the conclusion. The cardinal bound is exact: $T = D_\lambda$ is ordinal definable and meets every class in λ members. This sharpness statement concerns the numerical bound, not the optimality of the large-cardinal assumption.

6.2 Small families with nondefinable members

Theorem 6.4 (Simultaneous traces for a small definable family). Assume $\text{CR}(\lambda, q)$. Let $\mathcal{T} \in \text{OD}_{V_\lambda; q}$ satisfy

$$\mathcal{T} \subseteq \mathcal{P}(D_\lambda), \quad |\mathcal{T}| < \lambda.$$

Then every $T \in \mathcal{T}$ satisfies $|T \cap q| \in \{0, \lambda\}$. The members T need not individually belong to $\text{OD}_{V_\lambda; q}$.

Proof. Define

$$\mathcal{A} = \left\{ \bigcup (T \cap q) : T \in \mathcal{T}, 0 < |T \cap q| < \lambda \right\}. \quad (6.2)$$

This definition makes $\mathcal{A}$ a member of $\text{OD}_{V_\lambda; q}$, and $|\mathcal{A}| \leq |\mathcal{T}| < \lambda$.

For each T contributing to (6.2), put $\chi_T = |T \cap q| < \lambda$. Every member of $T \cap q$ is a countably infinite cofinal subset of λ . Therefore

$$\left| \bigcup (T \cap q) \right| \leq \max(\aleph_0, \chi_T) < \lambda.$$

The union is cofinal because the trace is nonempty. Thus every member of $\mathcal{A}$ belongs to C_λ . Cofinal rigidity forces $\mathcal{A} = \emptyset$. It follows that no $T \in \mathcal{T}$ has a nonempty trace of size below λ ; Lemma 2.1 gives the stated alternatives. $\square$

Corollary 6.5 (A common large intersection). At an ultraexacting λ , choose q as in Theorem 5.3. Every nonempty $\text{OD}_{V_\lambda; q}$ family of fewer than λ complete sections satisfies

$$\forall T \in \mathcal{T} \quad |T \cap q| = \lambda. \quad (6.3)$$

In particular, no nonempty OD_{V_λ} family of fewer than λ λ -thin complete sections exists. Thus a countable definable family of nondefinable transversals is also excluded.

Proof. All the traces are nonempty by completeness, so apply Theorem 6.4. If every member were λ -thin, any member would contradict (6.3). $\square$

The proof deliberately does not take the union of all the unions in (6.2). Their sizes can be unbounded in λ . Instead, $\mathcal{A}$ remains a small *family* of short cofinal sets, precisely the object prohibited by cofinal rigidity.

6.3 Uniformly indexed partial sections

Corollary 6.6 (Low-rank and ordinal indexing). Assume $\text{CR}(\lambda, q)$. Let θ be any ordinal and let $F \in \text{OD}_{V_\lambda; q}$ be a function with domain $V_\lambda \times \theta$ and values in $\mathcal{P}(D_\lambda)$. Then

$$\forall x \in V_\lambda \ \forall \alpha < \theta \quad |F(x, \alpha) \cap q| \in \{0, \lambda\}.$$

Hence a uniformly OD_{V_λ} family of λ -thin partial sections, indexed by $V_\lambda \times \theta$, cannot meet every class of Q_λ .

Proof. For fixed $x \in V_\lambda$ and ordinal α , the value $F(x, \alpha)$ belongs to $\text{OD}_{V_\lambda; q}$: add x to the low-rank parameters and α to the ordinal parameters. Apply Theorem 6.1. If all traces of every partial section are $< \lambda$, all traces on this particular q must be empty. $\square$

There is no cardinality bound on the indexing ordinal θ . This result uses uniform definability, rather than assuming that an arbitrary enumeration of the family is preserved by an embedding.

7 Non-compression of quotient equivalence

Definition 7.1 (Homomorphisms of equivalence relations). Let E be an equivalence relation on D_λ . A function $f : D_\lambda \rightarrow D_\lambda$ is an E_0 -to- E homomorphism when

$$a E_0 b \implies f(a) E f(b).$$

No converse implication, injectivity, surjectivity, or regularity of f is required.

Theorem 7.2 (Simultaneous large target classes). Assume $\text{CR}(\lambda, q)$. Let $E \in \text{OD}_{V_\lambda; q}$ be an equivalence relation on D_λ , and let $\mathcal{F} \in \text{OD}_{V_\lambda; q}$ be a family of E_0 -to- E homomorphisms with $|\mathcal{F}| < \lambda$. Then

$$\forall f \in \mathcal{F} \ \forall a \in q \quad |[f(a)]_E| \geq \lambda. \quad (7.1)$$

Proof. For each $f \in \mathcal{F}$, the class $[f(a)]_E$ is independent of the choice of $a \in q$: any two such a are E_0 -equivalent, and f is a homomorphism. This observation avoids choosing a representative of q .

Consider

$$\mathcal{A} = \left\{ \bigcup [f(a)]_E : f \in \mathcal{F}, a \in q, |[f(a)]_E| < \lambda \right\}. \quad (7.2)$$

Although the defining expression quantifies over every $a \in q$, at most one set is contributed by each f . Thus $|\mathcal{A}| \leq |\mathcal{F}| < \lambda$. The set $\mathcal{A}$ belongs to $\text{OD}_{V_\lambda; q}$, since E and $\mathcal{F}$ do; their finitely many low-rank parameters can be combined.

If $O = [f(a)]_E$ contributes to (7.2), then O is nonempty, every member of O is a countable cofinal subset of λ , and $|O| < \lambda$. Consequently

$$\sup \bigcup O = \lambda, \quad \left| \bigcup O \right| \leq \max(\aleph_0, |O|) < \lambda.$$

Therefore $\mathcal{A} \subseteq C_\lambda$. Cofinal rigidity implies $\mathcal{A} = \emptyset$, which is exactly (7.1). $\square$

Corollary 7.3 (Conditional inconsistency of small-class compression). Suppose λ is ultraexacting and $E \in \text{OD}_{V_\lambda}$ is an equivalence relation on D_λ all of whose classes have cardinality below λ . There is no nonempty OD_{V_λ} family of fewer than λ E_0 -to- E homomorphisms. In particular, there is no such homomorphism in OD_{V_λ} .

Proof. Choose q from Theorem 5.3. A nonempty proposed family would contain some f ; for any $a \in q$, Theorem 7.2 would give $|[f(a)]_E| \geq \lambda$, contrary to the assumption. The singleton family $\{f\}$ gives the final assertion. $\square$

The target-class bound is pointwise: the sizes of the E -classes may be unbounded in λ . A uniform bound $|[b]_E| \leq \mu < \lambda$ is not required. Moreover the homomorphisms inside the family are not required to be individually definable.

Corollary 7.4 (Small-group orbit obstruction). Let a group Γ of cardinality below λ act on D_λ , and suppose its orbit equivalence relation is in OD_{V_λ} . At an ultraexacting λ , no nonempty OD_{V_λ} family of fewer than λ homomorphisms sends E_0 into that orbit relation.

Proof. For every $b \in D_\lambda$, the map $\gamma \mapsto \gamma \cdot b$ surjects from Γ onto its orbit. Thus each orbit has size at most $|\Gamma| < \lambda$, and Corollary 7.3 applies. $\square$

Remark 7.5 (Two necessary qualifications). The class-size threshold in Corollary 7.3 is sharp: $E = E_0$ has classes of size λ , and the identity map is an ordinal-definable homomorphism. We do *not* prove that the family-size lower bound λ is attained for every small-class target relation. Also, without definability, constant homomorphisms exist by choosing any one member of D_λ ; the obstruction is not an abstract absence of functions.

8 A direct preserved-enrichment inconsistency

This section gives a local negative result of the kind sought in the Icarus part of the research plan. Its extra preservation assumption is explicit. There is no claim that a canonical Icarus parameter must satisfy it.

Theorem 8.1 (No preserved thin complete section). Work in ambient ZFC. Let N be a transitive inner model of ZF containing every element of $V_{\lambda+1}$. Suppose

$$j : N \longrightarrow N$$

is elementary, has set-sized restrictions, satisfies $j(\lambda) = \lambda$, and has critical point below λ . There cannot be a λ -thin complete section $T \subseteq D_\lambda$ with

$$T \in N, \quad j(T) = T. \quad (8.1)$$

Thinness and cardinalities are evaluated in the ambient universe.

Proof. The restriction $e = j \upharpoonright V_\lambda$ is a full elementary self-embedding of V_λ . The proof of Lemma 3.1 applies externally, so its critical sequence is cofinal in λ . Its graph, and hence its range a , belong to $V_{\lambda+1} \subseteq N$. As before, $j(a) = a \setminus \{\text{crit}(j)\}$.

Because N contains every subset of λ , it computes D_λ and finite-difference equivalence correctly. In particular $q = [a]$ belongs to N and $j(q) = q$. If (8.1) held, the set

$$B = T \cap q$$

would belong to N , be fixed by j , be nonempty, and have ambient cardinality below λ . It consists of short cofinal sets.

We now apply the proof of Lemma 3.4 inside this transitive setting, keeping its cardinal estimates external. The spectrum $S = \{\text{ot}(u) : u \in B\}$ and its union constructions belong to N by ZF. Order types of sets of ordinals are absolute. If S is cofinal, its increasing enumeration lies in N and the proof of Lemma 3.3 yields a contradiction. If it is bounded, the same external cardinal-product bound makes $\bigcup B$ a short cofinal set fixed by j , giving the same contradiction. Choice in N has not been used. $\square$

Corollary 8.2 (Forbidden Icarus-type parameter). There is no elementary self-embedding

$$j : L(V_{\lambda+1}, T) \longrightarrow L(V_{\lambda+1}, T)$$

with $\text{crit}(j) < \lambda$, $j(\lambda) = \lambda$, and $j(T) = T$, when $T \subseteq D_\lambda$ is a λ -thin complete section in the ambient universe and the usual inner-model and set-restriction hypotheses of Theorem 8.1 hold.

Proof. Apply Theorem 8.1 to $N = L(V_{\lambda+1}, T)$. $\square$

For a parameter $T \subseteq D_\lambda \subseteq V_{\lambda+1}$, there is no need to pretend that a quotient class belongs to $V_{\lambda+1}$: the class is formed in the full inner model. The notation $L(V_{\lambda+1}, T)$ is being used in the standard sense with $V_{\lambda+1}$ adjoined and T available as a set or predicate; the preservation requirement remains a separate hypothesis.

Proposition 8.3 (Small preserved families). Under the embedding assumptions of Theorem 8.1, let $\mathcal{T} \in N$ satisfy $j(\mathcal{T}) = \mathcal{T}$ and $|\mathcal{T}|^V < \lambda$. Then, for the critical tail class q constructed in that proof,

$$\forall T \in \mathcal{T} \quad |T \cap q|^V \in \{0, \lambda\},$$

provided every $T \in \mathcal{T}$ is a subset of D_λ .

Proof. The external test $|T \cap q|^V < \lambda$ need not be correctly computed in N , so we do not use it as an internal defining predicate. Instead define in N

$$\mathcal{A} = \left\{ \bigcup (T \cap q) : T \in \mathcal{T}, T \cap q \neq \emptyset, \text{ot}\left(\bigcup (T \cap q)\right) < \lambda \right\}.$$

The order-type condition is absolute between N and V . Thus $\mathcal{A}$ is fixed by j , is a family of short cofinal sets, and has ambient size below λ . The transitive version of Lemma 3.4 in the preceding proof forces $\mathcal{A} = \emptyset$. Any nonempty $T \cap q$ of ambient size below λ would have contributed to $\mathcal{A}$, by the countable-union estimate used in Theorem 6.4. There are no such traces, and every trace is a subset of the λ -sized class q . $\square$

Remark 8.4 (Scope of the class-embedding assertion). The proper-class notation in this section can be read in an ambient class theory permitting the stated restrictions, or in the usual set-coded formulation appropriate to the particular I0-type hypothesis. No unrestricted quantification over class embeddings is being asserted as a single sentence of bare ZFC. By contrast, the earlier ultraexactness and definability obstructions are set-sized first-order statements.

9 Consistency calibration and explicit nonclaims

Define T_{rig} to be the first-order theory

$$\text{ZFC} + \exists \lambda \exists q \left[\text{UEX}(\lambda) \wedge V_\lambda \subseteq \text{HOD} \wedge q \in Q_\lambda \wedge \text{CR}(\lambda, q) \right]. \quad (9.1)$$

Ordinal definability in (5.2) is first-order definable, so this is genuinely a first-order addition to the chosen standard formulation of ultraexactness.

Theorem 9.1 (An I0-calibrated consistency result). Over the usual arithmetical metatheory for consistency statements,

$$\text{Con}(\text{ZFC} + \text{I0}) \iff \text{Con}(\mathsf{T}_{\text{rig}}). \quad (9.2)$$

Every model of T_{rig} satisfies the trace, small-family, and quotient-compression conclusions above at its indicated λ and q .

Proof. Assume $\text{Con}(\text{ZFC} + \text{I0})$. By Input **C**, the existence of an ultraexactness cardinal is consistent, and the cited forcing result preserves such a λ while making $V_\lambda \subseteq \text{HOD}$. In the resulting theory, Theorem 5.3 supplies q satisfying $\text{CR}(\lambda, q)$. This proves the forward consistency implication. This is an application of a formalizable relative-consistency result; no existence of a transitive model follows merely from a consistency assumption.

Conversely, T_{rig} implies that an ultraexactness cardinal exists. Input **C** then gives $\text{Con}(\text{ZFC} + \text{I0})$ from $\text{Con}(\mathsf{T}_{\text{rig}})$. The remaining assertions are the already proved consequences of $\text{CR}(\lambda, q)$. $\square$

The new information in Theorem 9.1 is the cofinal-rigid parameter and the associated strengthened obstructions in the calibrated model. The equivalence of the two *standard* large-cardinal consistency strengths is imported, not newly established here. Adding a proved consequence does not increase consistency strength.

There is no conflict with $V_\lambda \subseteq \text{HOD}$: the forbidden representatives have rank λ , not rank below λ . Agreement on every lower rank does not yield a definable representative of a cofinal tail class. Moreover, Choice in the ambient universe produces a transversal T of Q_λ , since Q_λ is a set of nonempty sets. Corollary 6.3 says that this T cannot be in OD_{V_λ} , and cannot be in $\text{OD}_{V_\lambda, q}$ for a cofinal-rigid q .

What has not been proved

No contradiction from ultraexactness alone is obtained. No inconsistency of I0 , no resolution of a cover-exacting cardinal above a strongly compact cardinal, and no solution of the cardinal-preserving global-embedding problem follows. The family-size result for transversals does not resolve the synthesis's separate finite-subset-selector question. Optimality of ultraexactness as a hypothesis is also not established.

10 Further questions and a usable test

The proof suggests a concrete test for a proposed enrichment of a rank-into-rank domain. Ask whether the enrichment canonically yields a nonempty family of short cofinal sets of ambient size below λ that is preserved by the embedding. A positive answer supplies a contradiction by Lemma 3.4, without a stationary partition or a finite-cycle argument. If the information instead yields a tail-invariant assignment, Corollary 5.4 is the corresponding definability test.

Question 10.1. Does ordinary exactingness suffice for the nonexistence of a λ -thin OD_{V_λ} complete section of D_λ/E_0 ? The local lemmas hold for exact witnesses, but the construction of $q \in X$ in Lemma 5.1 uses $j \upharpoonright V_\lambda \in X$.

Question 10.2. For a fixed small-class equivalence relation E on D_λ , what is the least size of a nonempty OD_{V_λ} family of E_0 -to- E homomorphisms? Theorem 7.2 gives a lower bound of λ ; it does not construct a family attaining it.

Question 10.3. Can one characterize the classes $q \in Q_\lambda$ satisfying $\text{CR}(\lambda, q)$, or obtain useful closure properties of the collection of such classes? The present proof constructs one from a sufficiently correct ultraexact witness but does not classify all of them.

Question 10.4. Does a natural, independently motivated Icarus parameter define a preserved λ -thin complete section or a small family of the kind excluded in Section 8? The obstruction is now explicit, but finding such a parameter is a separate mathematical task.

These questions separate the actual deductions from their potential use in the larger inconsistency program. In particular, the present argument gives a checkable obstruction to specific extra information, rather than a reason to discard a large-cardinal axiom on aesthetic grounds.

A Definability, reflection, and the parameter minimum

For completeness, here are the formal ingredients behind Section 4. Fix set parameters p, r . A definition code consists of an ordinal β , a first-order formula number, and a finite tuple of ordinal parameters

below β , with $p, r \in V_\beta$. A code succeeds when its formula has a unique solution in the set structure $(V_\beta, \in)$ with those parameters. Satisfaction for this *set* structure is uniformly first-order definable in the ambient universe.

A set is in $\text{OD}(p, r)$ exactly when it is the value of a successful code. In one direction, a successful code gives a definition in V using β as one additional ordinal parameter. In the other, reflection applied to a particular definition in V supplies a sufficiently high V_β in which the same set is the unique solution. This does not use a truth predicate for the universe.

Order the codes by their height β , and within each height by a fixed well-order of formula numbers and finite ordinal tuples. Every initial segment is a set. Assign each definable object its least successful code, and compare objects by these least codes. This gives a uniformly first-order-definable, set-like well-order of $\text{OD}(p, r)$. Every nonempty definable subclass has a least member: below the code of one witness there is only a set of candidates, to which Separation and ordinary well-ordering apply.

The formulas used in the parameter lemma are now fixed formulas of set theory: the definition of $\text{Good}_\lambda(A)$; the assertion that some successful code with parameters p, r produces such an A ; and the assertion that A is least in the resulting well-order. Their finite logical complexity is bounded by some natural number m . Choosing $V_\zeta <_{\Sigma_m} V$ with m large enough makes each of them absolute for all relevant parameters in V_ζ . We do not need to know a numerically minimal m .

This uniform reflection is the reason the critical class q may be constructed *after* the height is chosen. The same formulas are correct for every $q \in V_\zeta$ that subsequently arises. An arbitrary ordinal used by an original definition might lie above the critical point, or even outside X ; no such ordinal is carried unchanged through the embedding. The least-counterexample formula replaces the original presentation.

The passage from a fixed parameter set $I \subseteq V_\lambda$ to a fixed individual parameter is similarly exact. The fixed ordinal

$$\rho = \min\{\text{rank}(p) : p \in I\}$$

lies below κ by normalization. Since κ is a limit ordinal, every possible witness of rank ρ lies in V_κ , and therefore in X , and is fixed. This is an external choice of a member of an already fixed low-rank stratum, not a hidden assertion of a definable well-order of I .

B Rank bookkeeping and proof audit

B.1 Ranks and domains

The following bounds use ordinary set-theoretic coding; increasing the height by a fixed finite amount would also suffice. Here $a \in D_\lambda$, $q = [a]$, and B is a nonempty subfamily of D_λ .

Object	Rank / location	Where the proof uses it
a , a short cofinal u	Rank λ ; in $V_{\lambda+1}$.	May be acted on by a full rank- $\lambda + 1$ embedding.
Critical sequence c	Rank λ with the standard graph code.	In an ultraexact X by definable recursion from $e \in X$; in N by $V_{\lambda+1} \subseteq N$.
$q, B, D_\lambda, C_\lambda$	Rank $\lambda + 1$ when nonempty.	Not elements of $V_{\lambda+1}$. They are handled in $X < V_\zeta$ or in N .
Q_λ ; a family $A \subseteq C_\lambda$	$\text{rank}(Q_\lambda) = \lambda + 2$; $\text{rank}(A) \leq \lambda + 1$.	All belong to sufficiently high V_ζ . Membership in X is separately proved.
Maps and families of maps	Finite rank overhead above the objects coded.	Heights are chosen above $\lambda + \omega$ and sufficiently correct.

A set's low rank does not by itself imply that it belongs to the nontransitive submodel X once its rank reaches λ . The proof establishes membership using unique definitions, except for the ultraexact restriction e , whose membership is an explicit hypothesis. In particular, it never applies a map with domain only $V_{\lambda+1}$ to the whole quotient Q_λ .

B.2 Dependency ledger

Deduction	Dependencies
Normalization	Local Kunen inconsistency; exact witness.
No fixed short cofinal set	Normalization; unique increasing enumeration.
No small fixed family	Previous lemma; fixed order-type spectrum; one bounded cardinal product.
Fixed-parameter rigidity	Small-fixed-family lemma; first-order $\text{OD}(p, r)$ well-order; uniform reflection; minimal parameter rank.
Cofinal-rigid class	Fixed-parameter rigidity; ultraexactness puts the critical sequence in X .
Trace and multisection barriers	Cofinal rigidity; each E_0 -class has exactly λ members.
Small-family and homomorphism barriers	Cofinal rigidity; a $< \lambda$ family of countable sets has a short union; no union over all resulting short sets.
Preserved-enrichment obstruction	Local small-fixed-family proof in a transitive N ; external Choice only for size estimates.
Consistency calibration	New cofinal-rigidity theorem plus the imported I0 comparison and coding preservation.

B.3 Checks against common invalid shortcuts

A singular cardinal is never treated as regular. All union estimates either concern a uniformly countable family, or follow after the order-type spectrum has been shown bounded. The potentially unbounded family in (6.2) is not flattened.

A fixed family is not assumed pointwise fixed. Lemma 3.4 works with its invariant spectrum and union. There is no unproved extension of a finite-permutation argument to an infinite family.

Ordinal parameters are not assumed fixed. The selected least counterexample is definable from

parameters that have actually been shown fixed. The uniform parameter lemma also explains why a single q handles every low-rank parameter.

The critical sequence is internal only when justified. In Section 5 this comes from $e \in X$. In Section 8 it comes from the full inclusion $V_{\lambda+1} \subseteq N$. Ordinary exactingness alone is not substituted for either condition.

The inner model's cardinal arithmetic is not imported into the ambient estimate. Proposition 8.3 uses an absolute order-type condition rather than an internal test of external smallness. No assumption that N satisfies Choice is needed.

Relative consistency is not transitive-model existence. Theorem 9.1 compares formal consistency assertions via imported formalizable results. It does not infer a transitive model from $\text{Con}(\text{ZFC} + \text{I0})$.

Sharpness is stated only where witnessed. The family q witnesses (5.3); $T = D_\lambda$ witnesses the exact trace bound; $E = E_0$ with the identity witnesses the target-class threshold. No matching construction for every family-size lower bound is asserted.

C Relation to the supplied reports and literature

References [1, 2] identify the two supplied source files. The comparison concerns the synthesis's cofinal-map-width, finite-choice, and transversal results; its other claims are not proof inputs here.

Critical-sequence normalization and fixed-cofinal-set arguments are established ingredients. Goldberg's July 2026 notes, reporting joint work with Blue, use an even/odd finite-difference orbit to obstruct a linear order in Remark 2.4 [7]. Here the additional ingredients are a fixed whole tail class, the small-fixed-family lemma, and simultaneous parameter minimization.

The checked primary sources are the versions of [3, 4, 6, 7] identified below. The parameter-relative simultaneous theorem extends the supplied synthesis. A targeted literature search does not establish published priority.

References

- [1] *Large cardinals at the inconsistency frontier: Unified Report*. User-supplied research plan, Large_Cardinals_Unified_Report.tex, in Cardinals2.zip, supplied 18 September 2026.
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- [3] Juan P. Aguilera, Joan Bagaria, and Philipp Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568v4](https://arxiv.org/abs/2411.11568v4), version of 16 September 2025.
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- [6] Joel David Hamkins, Greg Kirmayer, and Norman Lewis Perlmutter. Generalizations of the Kunen inconsistency. *Annals of Pure and Applied Logic* **163**(12) (2012), 1872–1890. [arXiv:1106.1951v2](https://arxiv.org/abs/1106.1951v2). See especially the discussion of the set-sized $V_{\mu+2}$ form in Section 1.
- [7] Gabriel Goldberg, reporting joint work with Doug Blue. *Consistency beyond the Kunen inconsistency*. Notes dated 1 July 2026, especially Remark 2.4. <https://math.berkeley.edu/~goldberg/Slides/CoverExact.pdf>.